

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15A.1

CHIRALITY AND ENANTIOMERS

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: ORGANIC CHEMISTRY, CARBONYLS AND CHIRALITY
SUB-TOPIC:	15A.1: Chirality and Enantiomers
TOTAL QUESTIONS:	4 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	6 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15A.1

Q1. 1. (a) Explain what is meant by a chiral centre in an organic molecule.

[1 Marks]

WCH14/01/Jan23/Q4(a)

Q2. (b) Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, displays optical isomerism. Draw 3D wedge-and-dash diagrams of the two enantiomers of lactic acid.

[2 Marks]

WCH14/01/Jan23/Q4(b)

Q3. (c) Explain why 2-chlorobutane is chiral whereas 1-chlorobutane is achiral.

[2 Marks]

WCH14/01/Oct22/Q3(c)

Q4. 2. MCQ: Which of the following molecules exists as a pair of optical isomers?

[1 Marks]

WCH14/01/Jun22/Q1

- A. Propan-2-ol
- B. Butan-2-ol
- C. Pentan-3-ol
- D. 2-methylpropan-2-ol

SECTION B: WORKED MARK SCHEME — 15A.1

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Explain what is meant by a chiral centre in an organic molecule. A carbon atom attached to four different groups/atoms.	[1]
Q2	Q2: (b) Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, displays optical isomerism. Draw 3D wedge-and-dash diagrams of the two enantiomers of lactic acid. 1. Correct 3D tetrahedral arrangement around central carbon. 2. Non-superimposable mirror images clearly shown.	[2]
Q3	Q3: (c) Explain why 2-chlorobutane is chiral whereas 1-chlorobutane is achiral. 1. 2-chlorobutane has C2 bonded to -H, -Cl, -CH ₃ , -CH ₂ CH ₃ (4 different groups). 2. 1-chlorobutane C1 has two -H atoms attached (not 4 different groups).	[2]
Q4	Q4: 2. MCQ: Which of the following molecules exists as a pair of optical isomers? B (Butan-2-ol has C2 attached to H, OH, CH ₃ , C ₂ H ₅).	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15A.1**FAQ #1: How do I identify a chiral carbon in a skeletal formula?****Category: Structural Recognition****COMMON EXAMINER TRAP / MISTAKE:**

Forgetting implicit hydrogen atoms attached to carbons in skeletal structures.

MODEL ANSWER / EXAMINER EXPECTATION:

Count all four bonds from the carbon node. Include any implicit H atom. If all 4 attached groups are distinct, it is a chiral centre.

FAQ #2: Are enantiomers chemically identical?**Category: Physical & Chemical Properties****COMMON EXAMINER TRAP / MISTAKE:**

Stating enantiomers have different boiling points or chemical reactions with achiral reagents.

MODEL ANSWER / EXAMINER EXPECTATION:

Enantiomers have identical physical properties (b.p., m.p., solubility) and chemical reactions with achiral reagents. They ONLY differ in their rotation of plane-polarised light and reactions with chiral reagents/enzymes.